JACOB RAUSCHER

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EDUCATION

University of South Florida, ***Tampa, FL***

Accelerated Master of Science in Electrical Engineering EXPECTED Dec. 2027

Concentration in Automation and Control Systems

- Relevant Coursework: Machine Learning, Deep Learning, Embedded Systems, Applied Mechatronics, Applied Optimization, Applied Industrial AI

Bachelor of Science in Electrical Engineering, Summa Cum Laude May 2026 | GPA: 3.92

- Relevant Coursework: Programming with C, Computer Aided Design & Engr, Electrical Systems I & II, Linear Controls, Signals & Systems, Embedded Systems, Mechatronics Systems, Microprocessor Principles

PROFESSIONAL EXPERIENCE

Engineering Intern | Schweitzer Engineering Labs (SEL) Engineering Services Apr. 2026 - Present

- Configured reclosing logic on an SEL-700G generator protection relay for a TECO project, translating protection requirements into field-ready settings
- Modeled inverter-based resource (IBR) sites and compared response to Ride-Through conditions to test PRC-29 standards, and authored technical reports documenting performance results and design findings
- Produced and reviewed CAD drafting deliverables for accuracy and design intent, coordinating across multiple engineering teams to meet strict project deadlines

Undergraduate Researcher, Design Lead | RARE Lab | University of South Florida Aug. 2025 - Jan. 2026

- Led mechanical design of custom mounting hardware for an Augmented Reality Fog Screen Projector on a Unitree GO2 robotic dog, designing, 3D printing, and assembling all parts in SolidWorks
- Independently researched electronics and hardware requirements for the platform and delivered reliable mechanical solutions

PROJECTS

MSEE Thesis: STAR (Smart Telemetric Action Ring) Aug. 2026 - Present

- Designing a ring-sized wearable gesture controller with an IMU, Bluetooth LE, haptic feedback, and Qi wireless charging for smart device control
- Developing a custom machine learning gesture classification algorithm on raw IMU time-series data, letting users define custom gestures from minimal training examples

SplatMotion: Senior Design Project for ISO Technologies Inc. Aug. 2025 - May 2026

- Developed with a 4-person team, an end-to-end smartphone-based 3D limb scanning system for a sponsor, taking a raw Android capture through cloud reconstruction to a viewable, exportable mesh
- Architected a computer vision pipeline: ARCore-enabled Android capture (Kotlin), SfM camera pose estimation, 3D Gaussian Splatting, SuGaR surface extraction, and Ball-Pivoting Algorithm meshing to a .OBJ output
- Debugged and optimized the pipeline by isolating checkpoint corruption, phantom mesh geometry, and coordinate frame mismatches, and tuned reconstruction parameters to improve mesh fidelity

Analog Wah Pedal: IEEE Expo Jan. 2026 - May 2026

- Designed, built, and tested an analog Wah effects pedal from scratch, prototyping a swept band-pass filter on a breadboard and tuning the circuit for the desired center frequency range and resonance (Q)
- Modeled and validated the filter circuit with oscilloscope and function generator measurements, characterizing frequency response and signal integrity across the audio band, and then presented at IEEE Expo

SKILLS

Programming: Python, C/C++, MATLAB, Simulink, RSLogix 5000 (Ladder Logic), JavaScript, HTML
Engineering & CAD: SolidWorks, AutoCAD, Altium Designer, PCB Design, 3D Printing
Embedded Systems & Hardware: MSP430, ESP32, Arduino, Microcontrollers, Soldering, Breadboarding
Lab & Controls: Oscilloscopes, Function Generators, Multimeters, PID Tuning, PSpice, LTspice
Certifications: SolidWorks, MATLAB, Simulink, Microsoft Office Specialist